

Homework 3: Policy Gradient Methods

CS 1840/Stat 184: Introduction to Reinforcement Learning

(Due November 11 at 11:59 pm ET)

0 Instructions

1. Under each problem, please list all your collaborators and reference materials (beyond those specified on the website).
2. Submit your report as a *single* pdf file on Gradescope under the assignment “HW3” only. Please tag your pages correctly as is requested in Gradescope.
3. We highly encourage you to use $\text{\LaTeX}$. We will also accept neatly written handwritten homework.
4. Please see the separate programming assignment on Canvas for submission instructions there.

1 PG: Alternative Expressions and Baselines [30 points]

In this question, you will walk through a series of ways to express the policy gradient. Though they all equal the same *value*, they correspond to different *gradient estimators* with different *variances*. Generally speaking, gradient estimators with lower variances lead to more stable improvement when used for gradient ascent (or descent).

Consider a finite horizon Markov Decision Process $\mathcal{M} = \{S, \mathcal{A}, H, r, P, \mu_0\}$. Let π_θ be a parameterized, stochastic policy, that is, $\pi_\theta(a | s) \in [0, 1]$ is the probability of choosing action a in state s , where $\theta \in \mathbb{R}^d$ are the policy parameters.

Denote the expected cumulative reward by

$$J(\theta) = \mathbb{E}_{\tau \sim \rho_\theta} [R(\tau)]$$

where $R(\tau) = \sum_{h=0}^{H-1} r(s_h, a_h)$ is the total reward and the density over trajectories $\tau = (s_0, a_0, s_1, \dots, s_{H-1}, a_{H-1})$ is given by acting in the environment according to the policy:

$$\rho_\theta(\tau) = \mu_0(s_0) \pi_\theta(a_0 | s_0) P(s_1 | s_0, a_0) \pi_\theta(a_1 | s_1) \dots P(s_{H-1} | s_{H-2}, a_{H-2}) \pi_\theta(a_{H-1} | s_{H-1}).$$

Recall that $\nabla_\theta J(\theta) \in \mathbb{R}^d$ is called the **policy gradient**. We can optimize a parameterized policy by incrementing the policy parameters θ in the direction of the policy gradient. We proved in class that the policy gradient can be expressed as

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \rho_\theta} \left[\sum_{h=0}^{H-1} \nabla_\theta \log \pi_\theta(a_h | s_h) \cdot R(\tau) \right].$$

In this problem, we complete the proof of a few claims we made in class.

1. [5 points] Show that for any fixed timestep h ,

$$\mathbb{E}_{\tau \sim \rho_\theta} \left[\nabla_\theta \log \pi_\theta(a_h | s_h) \cdot \sum_{k=0}^{h-1} r(s_k, a_k) \right] = 0.$$

This captures the intuition that the action taken at time h cannot influence any of the rewards that come *before* it.

2. [5 points] Using the above, prove that the following expression holds:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \rho_\theta} \left[\sum_{h=0}^{H-1} \left(\nabla_\theta \log \pi_\theta(a_h | s_h) \cdot \sum_{k=h}^{H-1} r(s_k, a_k) \right) \right].$$

This expression should appear intuitive: taking action a_h can only influence rewards that come after it, so we should only make taking a_h more likely if the rewards *following* it are high.

3. **[10 points]** Now, using the previous expression, show that the following expression holds:

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim \rho_{\theta}} \left[\sum_{h=0}^{H-1} \nabla_{\theta} \log \pi_{\theta}(a_h | s_h) \cdot Q_h^{\pi_{\theta}}(s_h, a_h) \right].$$

This is the most common expression stated for the functional form of a policy gradient (in the finite-horizon setting).

4. **[10 points]** Now we will use **baseline** functions to reduce the variance. Let $b_h : S \rightarrow \mathbb{R}$ for $h \in \{0, \dots, H-1\}$ be a sequence of baseline functions. Show that

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim \rho_{\theta}} \left[\sum_{h=0}^{H-1} \nabla_{\theta} \log \pi_{\theta}(a_h | s_h) \cdot (Q_h^{\pi_{\theta}}(s_h, a_h) - b_h(s_h)) \right].$$

This form is commonly used in practice, as the use of a baseline $b_h(s)$ helps to reduce the variance in the gradient estimate.

2 Policy Gradient Examples [30 points]

We will now derive the policy gradient expression for two widely used policy classes.

Softmax Policy with a Linear Model

Let us consider the following *softmax policy*. Given some feature function $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$, the softmax policy is the stochastic, parameterized policy that assigns probability to actions according to

$$\pi_{\theta}(a \mid s) \propto \exp(\theta^\top \phi(s, a))$$

where $\theta \in \mathbb{R}^d$ are the policy parameters.

1. [3 points] Consider a finite MDP with $\mathcal{S} = \{1, 2, \dots, S\}$ and $\mathcal{A} = \{1, 2, \dots, A\}$. Let $d = SA$ so that each entry of θ corresponds to a state-action pair. Let $\theta_{s,a}$ be the entry of θ corresponding to the state-action pair (s, a) . Write a feature function ϕ such that $\pi_{\theta}(a \mid s) \propto \exp(\theta_{s,a})$.
2. [3 points] Intuitively, the softmax policy is more likely to choose actions for which the feature vector $\phi(s, a)$ is more aligned with the parameter θ . Prove that

$$\pi_{\theta}(a \mid s) \geq \pi_{\theta}(a' \mid s) \quad \text{if and only if} \quad \theta^\top \phi(s, a) \geq \theta^\top \phi(s, a')$$

3. [6 points] Prove that

$$\nabla_{\theta} \log \pi_{\theta}(a \mid s) = \phi(s, a) - \mathbb{E}_{a' \sim \pi_{\theta}(\cdot \mid s)} [\phi(s, a')]$$

4. [6 points] Using this expression, show that:

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim \rho_{\theta}} \left[\sum_{h=0}^{H-1} A_h^{\pi_{\theta}}(s_h, a_h) \phi(s_h, a_h) \right]$$

Gaussian Policy with a Linear Model

Let us examine stochastic policies for *continuous* action spaces $\mathcal{A} = \mathbb{R}^k$. We consider the class of policies that output an action normally distributed around the output of a linear model. Concretely, given some feature mapping $\phi : \mathcal{S} \rightarrow \mathbb{R}^d$ and parameter matrix $\Theta \in \mathbb{R}^{k \times d}$, we define the distribution over actions as

$$\pi_{\Theta}(\cdot \mid s) = \mathcal{N}(\Theta \phi(s), \sigma^2 \mathbf{I}_k),$$

where $\mathbf{I}_k$ is the $k \times k$ identity matrix and $\sigma^2 > 0$ is a constant that specifies the distribution's variance. The probability density function over actions is then given by

$$\pi_{\Theta}(a \mid s) = \frac{1}{\sqrt{2\pi k}} \exp\left(-\frac{1}{2\sigma^2} \|a - \Theta \phi(s)\|^2\right).$$

5. **[6 points]** Using matrix notation, derive an expression for $\nabla_{\Theta} \log(\pi(\mathbf{a} \mid s))$. (For a refresher on vector calculus rules, consider checking out the [Matrix Cookbook](#). You can also derive these rules by considering one element at a time.)
6. **[6 points]** Using this expression, show that:

$$\nabla_{\Theta} J(\Theta) = \frac{1}{\sigma^2} \mathbb{E}_{\tau \sim \rho_{\Theta}} \left[\sum_{h=0}^{H-1} A_h^{\pi_{\Theta}}(s_h, \mathbf{a}_h) \mathbf{a}_h \phi(s_h)^{\top} \right]$$

3 Fitted DP [20 points]

Recall the **fitted DP** algorithm, which uses *supervised learning* to approximate the Q-function. In lecture, we explored this problem in the finite horizon case. Here we will extend it to the infinite-horizon case, where it is impossible to store a dataset of trajectories on a computer (since a trajectory has infinite length). To get around this issue, we modify the rollout procedure: at each timestep, we terminate the trajectory with probability $1 - \gamma$. Notice that this strategy yields finite trajectories, but also causes the length of each trajectory to be itself stochastic.

Expected Value of Truncated Rewards

Call the number of (s, a) pairs in a trajectory collected this way $C \sim \text{Geom}(1 - \gamma)$ —in particular, note that $P(C \geq c) = \gamma^c$. We seek to show that the expected value of the rewards collected in this manner allows us to recover the expected infinite-horizon discounted reward.

In particular:

1. [6 Points] Show that

$$\mathbb{E}_{\tau \sim \rho_0, C \sim \text{Geom}(1-\gamma)} \left[\sum_{h=0}^{\infty} \mathbb{I}\{C \geq h\} r(s_h, a_h) \right] = \mathbb{E}_{\tau \sim \rho_0} \left[\sum_{h=0}^{\infty} \gamma^h r(s_h, a_h) \right],$$

where ρ denotes a distribution of infinite-length trajectories under some fixed (stationary) policy. Thus, we see that the expected reward of the truncated trajectory is equal to the expected discounted reward in the infinite horizon case.

Using this fact, it follows (and you can take as given) that

$$V_C^\pi(s) = V^\pi(s),$$

where

$$V_C^\pi(s) = \mathbb{E}_{\tau \sim \rho_0, C \sim \text{Geom}(1-\gamma)} \left[\sum_{h=0}^{\infty} \mathbb{I}\{C \geq h\} r(s_h, a_h) \mid s_0 = s, \pi \right]$$

and V^π is the infinite-horizon discounted value, and likewise $Q_C^\pi(s, a) = Q^\pi(s, a)$.

2. [4 Points] Conclude that, using this sampling scheme, we find the following Bellman consistency equation:

$$Q_C^\pi(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [V_C^\pi(s')].$$

Fitted Value Iteration

Recall the algorithm for fitted value iteration:

Input: offline dataset $\tau_1, \dots, \tau_N \sim \rho_{\pi_{\text{data}}}$

1. Initialize fitted Q function at f^0
2. For $k = 1, \dots, K$:

$$f^k = \arg \min_{f \in \mathcal{F}} \sum_{i=1}^N \sum_{h=1}^{H-1} \left(f(s_h^i, a_h^i, h) - \left(r(s_h^i, a_h^i) + \max_a f^{k-1}(s_{h+1}^i, a, h+1) \right) \right)^2$$

3. Treating f^K as an estimate of Q^* , return $\pi_h(s) = \arg \max_a \{f^K(s, a, h)\}$.

The question:

3. **[5 points]** Using the results above, extend the optimization in step 2 to the infinite horizon case, where we continue to use the sampling scheme above and let f^k approximate Q_C^* .

Fitted Policy Evaluation

Recall the algorithm for fitted policy evaluation in the finite-horizon setting:

Input: offline dataset $\tau_1, \dots, \tau_N \sim \rho_{\pi_{\text{data}}}$, deterministic policy π to evaluate

1. Let f^0 be the initial estimate of Q^*
2. For $k = 1, \dots, K$:

$$f^k = \arg \min_{f \in \mathcal{F}} \sum_{i=1}^N \sum_{h=0}^{H-1} \left(f(s_h^i, a_h^i, h) - \left(r(s_h^i, a_h^i) + f^{k-1}(s_{h+1}^i, \pi(s_{h+1}^i), h+1) \right) \right)^2$$

3. With f^K as an estimate of Q^* , return $\pi_h(s) = \arg \max_a \{f^K(s, a, h)\}$

The question:

4. **[5 points]** Extend the equation in step 2 to the infinite horizon case.